

Hybrid randomized clinical trials incorporating external controls: assumptions and related recommendations

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<https://github.com/foucher-y/talks>

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Augmenting randomized clinical trials (RCT) with external controls

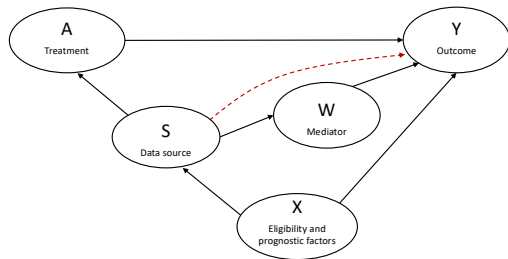
- ▶ RCTs represent the reference protocol for treatment evaluations.
- ▶ For large sample size, the comparability of the arms and the power acceptable.
- ▶ Number of trials are based on small sample sizes (median at 75 patients in ICU).¹
- ▶ Novel methods aim to increase the power by using internal and external controls.
- ▶ It calls for trade-off between:
 - ▶ decreasing the random bias, and
 - ▶ increasing systematic bias.

How to reduce the risk of systematic bias?

¹Anthon et al. (Journal of Clinical Epidemiology, 2019)

Notations and directed acyclic graph²

- ▶ $Y = (0, 1)$: the binary outcome.
- ▶ A : the treatment arm.
 - ▶ $A = 1$ for the experimental group.
 - ▶ $A = 0$ for the control group.
- ▶ S : the data source.
 - ▶ $S = 1$ for the internal RCT.
 - ▶ $S = 0$ for the external population.
- ▶ X : the causes of both Y and S .
- ▶ W : the mediators of being included in the RCT.



²Dang et al. (arXiv, 2023)

Estimand and targeted population

- ▶ Our objective is to estimate the average treatment effect (ATE) in the target population of the RCT.
- ▶ What would have been obtained from the RCT with a large sample size without augmentation with external controls?

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)|S = 1]$$

- ▶ $Y(0)$ is the counterfactual outcomes under the control treatment.
 - ▶ $Y(1)$ is the counterfactual outcomes under the experimental treatment.
- ▶ When **three assumptions for causal identifiability** are met, one can estimate the ATE from observations:³

$$\text{ATE} = \mathbb{E}[Y|A = 1] - \mathbb{E}_{X|S=1}[E(Y|A = 0, X)]$$

³Valancius (Biometrics, 2024)

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Assumption #1

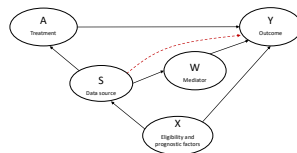
- ▶ Positivity violation occurs when it is certain that patients are allocated to a treatment group.
- ▶ Because of the randomization, this condition holds for individuals in the RCT.
- ▶ Therefore, positivity in treatment allocation is respected if the external population have a non-zero probability to be included in the RCT.

Positivity in RCT-recruitment (A1):

The external population have a non-zero probability to be included in the RCT.

Assumption #2

- ▶ The factors X may represent a source of bias.
- ▶ For instance, elderly patients are often under-represented in a RCT. It may lead to an overestimated effect of the experimental treatment.
- ▶ It calls for the absence of measuring X for correcting the bias by modelling.

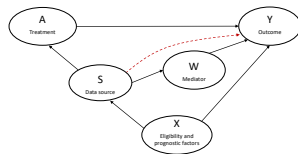


No unmeasured confounders (A2)

The eligibility criteria and prognostic factors X are measured.

Assumption #3

- ▶ The average outcome for controls given factors X is the same in the RCT and external population.
- ▶ It can be violated for different schedules of visits in a trial.
- ▶ The follow-up is an example of a mediator W .
- ▶ The measurement of W can further help in correcting the related bias.



No direct effect on the outcome of being in the trial (A3)

There is no residual effect of S on Y conditional on the confounders X and mediators W (who must be measured).

Recommendations to prevent violation of the assumptions by design

Assumption 1 Positivity in RCT-recruitment	1.1. Inclusion and exclusion criteria of the external and internal subjects are similar. 1.2. External controls come from a concomitant or recent study. 1.3. The external study consists for a large part of the same or similar centers.
Assumption 2 No unmeasured eligibility and prognostic factors X	2.1. Factors X are available in both the internal and external data. 2.2. Factors X are measured similarly across the internal and external studies. 2.3. Factors X should include social and lifestyle dimensions.
Assumption 3 No direct effect on the outcome	3.1. Baseline of the controls is identical in the RCT and the external study. 3.2. External controls receive the same treatment as in the control ones. 3.3. Patients in the RCT have a follow-up as close as possible as the routine care. 3.4. Outcome Y are measured similarly in the RCT and the external study. 3.5. Mediators W are measured similarly in the RCT and the external study. 3.6. Missing data on the variables (X, W, Y) is rare and (completely) at random.

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Treatment regression for evaluating the positivity in recruitment (A1)

- ▶ Description to check the overlap of the characteristics X between the internal and external controls.
- ▶ Estimation of a logistic regression (or another predictive model) with S as the outcome conditional on X to predict the individual probabilities to be included in the external or internal study.
- ▶ Identify the external patients with a probability lower than ϵ , for instance $\epsilon = 0.05$.

In case of positivity violation, the investigators can update the inclusion criteria of the external population to recover positivity in RCT-recruitment.

Outcome regression for evaluating the absence of direct effect of being in the RCT (A3)

- ▶ The respect of A3 results in similar outcomes for controls according to X and W regardless the source:
- ▶ In our context with a binary outcome, one can propose checking that $O/E = 1$:
 - ▶ O is the number of observed events among the internal controls, and
 - ▶ E is the expected number with the trained model from the external controls.
- ▶ Let N_{0c} the required sample size for such a calibration.⁴
- ▶ One can compare N_{0c} with N_{0s} , i.e. the required sample size of controls for an usual 1:1 superiority RCT.

⁴Riley et al. (Statistics in Medicine, 2021)

Two illustrative examples with $(\alpha, \beta) = (0.05, 0.20)$

▶ Example # 1:

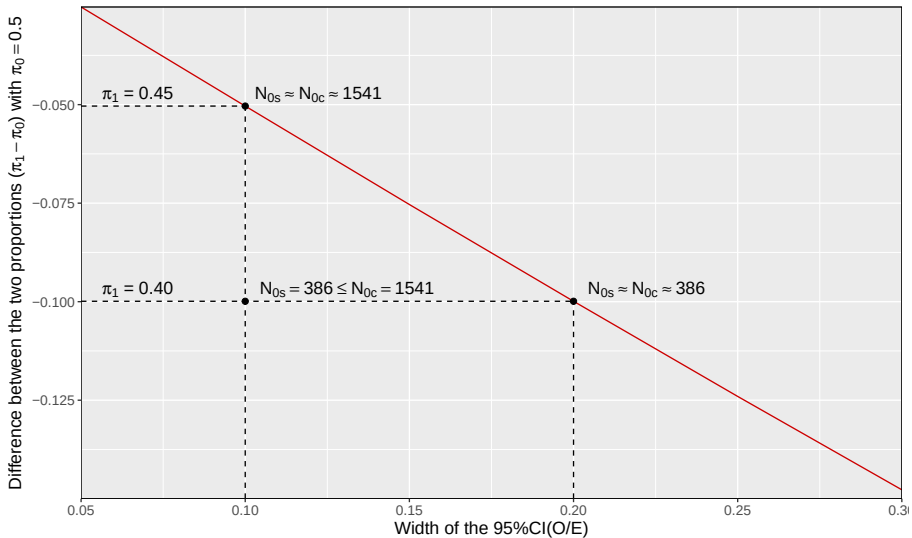
- ▶ Expected proportions of event are 0.50 and 0.45 in the two arms.
- ▶ Width of the CI of $(\log(O/E))$ equals to 0.20
- ▶ Sample size of controls for the usual trial: $N_{0s} \approx 1562$.
- ▶ Sample size of controls for calibration: $N_{0c} \approx 386$.
- ▶ **The effective of internal controls will allow for investing the calibration.**

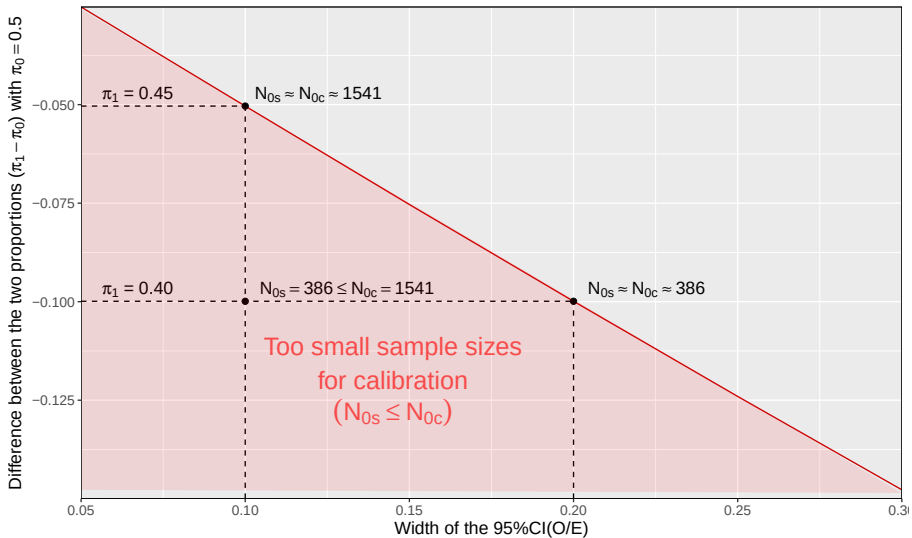
▶ Example # 2:

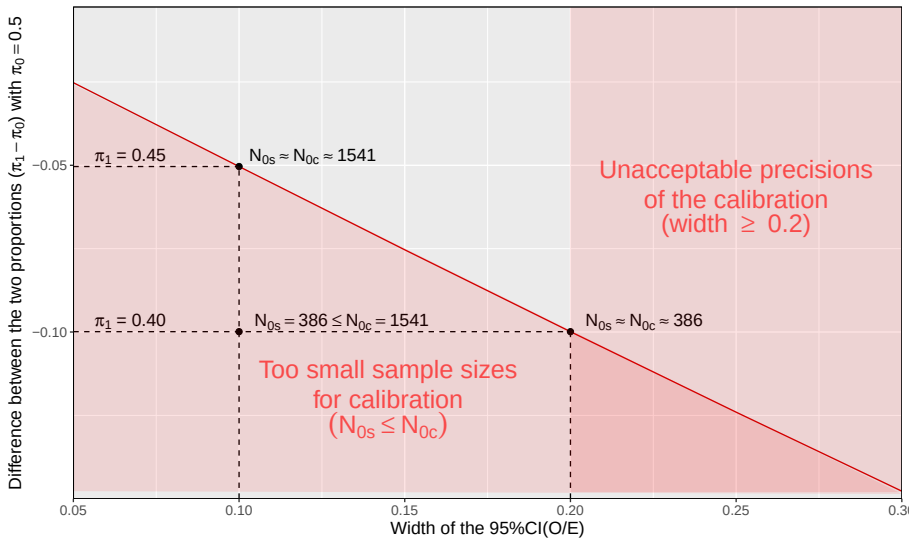
- ▶ Expected proportions of event are 0.50 and 0.35 in the two arms.
- ▶ Sample size of controls for the usual 1:1 superiority trial: $N_{0s} \approx 170$ patients.
- ▶ **The effective of internal controls will not allow for investing the calibration.**

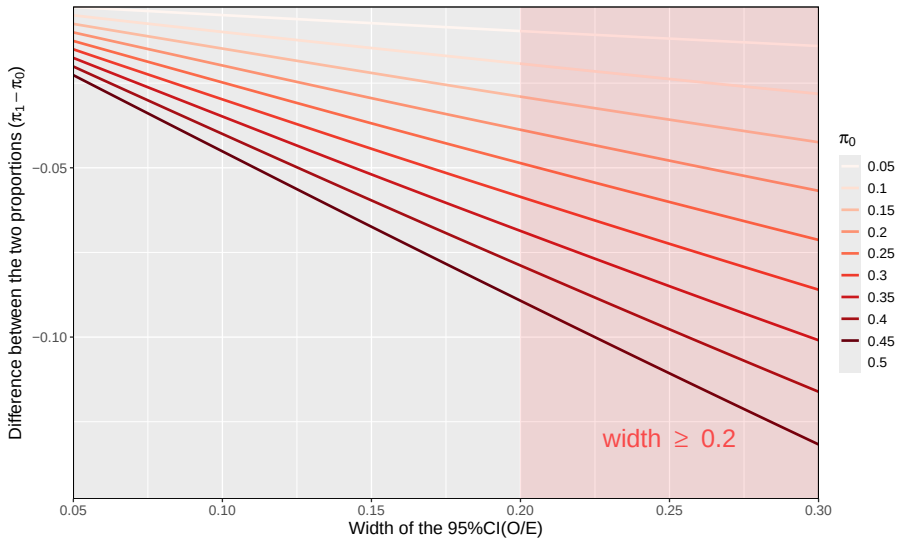
▶ Example # 3:

- ▶ Expected proportions of event are 0.50 and 0.40 in the two arms.









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Summary of the key points for such an hybrid trial

- ▶ External controls represent an opportunity, but 3 assumptions should be respected.
- ▶ The consideration of confounders (A2) is often discussed but represents only 1 assumption.
- ▶ We proposed a bundle of 12 recommendations which can guide the study design.
- ▶ It underlined the importance of prospectively designing both the internal and external sources as close as possible:
 - ▶ Retrospective registries or medico-administrative databases as external sources should be used with caution.
 - ▶ The prospective collection of the internal data must be designed according to the methods used for external controls.

References

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